

EDA HW

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$$1. \quad m(X) = \frac{1}{n} \sum_{i=1}^n a + bx_i \rightarrow \frac{1}{n} (Na + \sum_{i=1}^n bx_i) \\ = \frac{1}{n} (Na + b \sum_{i=1}^n x_i) \\ = a + b \left(\frac{1}{n} \sum_{i=1}^n x_i \right) \\ = a + b m(X)$$

$$2. \quad \text{Cov}(X, a+bY) = \frac{1}{n} \sum_{i=1}^n (x_i - m(X))(a + by_i - m(a+bY)) \\ = \frac{1}{n} \sum_{i=1}^n (x_i - m(X))(a + by_i - a - b m(Y)) \\ = \frac{1}{n} \sum_{i=1}^n (x_i - m(X))(by_i - b m(Y)) \\ = b \frac{1}{n} \sum_{i=1}^n (x_i - m(X))(y_i - m(Y)) \\ = b \cdot \text{Cov}(X, Y)$$

$$3. \quad \text{Cov}(a+bX, a+bX) = b \cdot \text{Cov}(X, a+bX) = b^2 \cdot \text{Cov}(X, X)$$

$$\text{Cov}(X, X) = \frac{1}{n} \sum_{i=1}^n (x_i - m(X))(x_i - m(X)) \\ = \frac{1}{n} \sum_{i=1}^n (x_i - m(X))^2 \rightarrow = s^2$$

4. Yes, the rank is the same when g is a non-decreasing transformation. The quantile is also the same in that the order is preserved. $IQR = a_{75} - a_{25}$ IQR of $g(X)$ is $g(a_{75}) - g(a_{25})$ and this is not equal to $g(IQR)$. The same applies to the range.

5. No, they are not equal. It is not true for non-decreasing functions.